magnet-log-collector — Service Overview

What it is, how it works, and its role in this project

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magnet-log-collector/ — Windows Service "MagnetLogCollector"
```

What it is

A small standalone Node.js background service — separate from the PrintApp web app — that runs on the magnet dimension-check measuring machine in the factory. It watches a folder for the daily result file exported by the measuring software and loads new rows into SQL Server.

It installs as a Windows Service named **MagnetLogCollector**, so it starts automatically with Windows and restarts itself if it crashes. No manual intervention needed day to day.

How it works

- Every 30 seconds (configurable), it looks for today's and yesterday's source file: `D:\MagnetLogfile_Summary\yyyyMMdd.csv` (falls back to `.xlsx` / `.xls` if that's what the machine exports).
- It copies the file to a temp folder before reading — if the measuring software still has the file locked, this cycle is simply skipped and retried next time, no error.
- It parses the CSV's 24 columns (barcode, test result, test time, unit, and 10 channels A–J of test value + polarity) and maps them into the destination table's columns. Rows missing a Serial Number, Status, or Date/Time are skipped and logged with the reason.
- **No offset/state file.** Every cycle re-checks the **whole** file against the database (by Serial Number + test time) and inserts only the rows not already there. This makes it self-healing: machine off overnight, database unreachable for a while, or the file rewritten — it always catches up on its own on the next cycle, without any special-case recovery logic.
- A built-in dashboard at `http://<machine>:8022` shows, per file, the total/valid/skipped row counts and whether the database is fully caught up (a CLI check command is also available for the same check without a browser).

Its role in this project

It writes directly into the same SQL Server database (`svn_pentaho`) that PrintApp uses — table `SVN_MiddleDimensionCheckResult` . It never talks to PrintApp directly; the database is the only thing connecting them.

> Magnet measuring machine (CSV export) → **magnet-log-collector** → SQL Server (`SVN_MiddleDimensionCheckResult`) → PrintApp Middle pages

That table is what powers two pages in PrintApp's Middle module:

- `/middle/result` — "Test Result from the Middle Panel Station": a searchable grid of every dimension-check result (filter by Serial Number, Pass/Fail, Work Order, date range).
- `/middle/summary` — "Work Order Summary": compares, per Work Order, how many units were dimension-tested against how many had production results entered, to catch mismatches (missing or extra serials).

In short: it is the bridge between the magnet measuring machine's raw export file and the QC dashboards operators use inside PrintApp.